

VPA IB DP CS — Topic 6: Resource Management (HL Only, Weeks 10 – 18)

Topic 6: Resource Management 资源管理 Zīyuán Guǎnlǐ HL ONLY

IB Syllabus: Topic 6 (6.1 – 6.2) | Level: HL ONLY | Duration: 9 weeks (30 hours)

Topic 6.1: Operating Systems (Advanced) 操作系统（高级）Cāozuò Xìtǒng (Gāojí)

Week 10 — CPU Scheduling CPU调度 CPU Diàodù

Lesson	Content	Activities	Differentiation
1	Processes vs threads; process control block (PCB); context switching (进程, 线程, 进程控制块)	Diagram PCB contents; trace context switch	Bronze: Identify PCB fields; Silver: Explain context switch; Gold: Compare process vs thread overhead
2	Scheduling algorithms: FCFS, SJF, Round Robin, Priority scheduling	Calculate waiting time and turnaround for each algorithm	Bronze: Trace FCFS; Silver: Compare SJF and Round Robin; Gold: Analyse starvation in priority scheduling
3	Preemptive vs non-preemptive scheduling; convoy effect; starvation	Scenario: choose algorithm for real-time system	All: Participate; Gold: Discuss multilevel feedback queues
4	Real-time scheduling: hard vs soft real-time; deadline scheduling	Case study: airbag system, streaming video	All: Classify; Gold: Design schedule for given tasks
5	TOK: CPU scheduling algorithms embody different notions of "fairness." Is fairness a mathematical property or a moral one? Can an algorithm be unfair?	Debate; written argument	All: Contribute; Gold: Reference Rawls' theory of justice

TOK Knowledge Question: CPU scheduling algorithms must balance efficiency, fairness, and responsiveness. When a scheduler prioritises one process over another, is this a technical decision or an ethical one? Can we objectively determine the "best" scheduling algorithm, or does "best" depend on what we value?

Week 11 — Memory Management 内存管理 Nèicún Guǎnlǐ

Lesson	Content	Activities
1	Memory management overview: partitioning, fragmentation (internal/external), compaction (分区, 碎片, 紧缩)	Calculate fragmentation; design compaction strategy

Lesson	Content	Activities
2	Paging: page tables, logical vs physical address, page faults, translation lookaside buffer (TLB)	Translate logical to physical addresses; calculate page table size
3	Segmentation: segment tables, base and limit registers; segmentation vs paging	Compare paging and segmentation with examples
4	Virtual memory revisited: demand paging, page replacement algorithms (FIFO, LRU, optimal)	Simulate page replacement; count page faults for each algorithm
5	Formative: Memory management calculations (25 marks)	Timed; peer marking

Week 12 — Deadlock and Concurrency 死锁与并发 Sǐsuǒ yǔ Bìngfā

Lesson	Content	Activities
1	Deadlock: conditions (mutual exclusion, hold and wait, no preemption, circular wait); resource allocation graphs	Identify deadlock in resource graphs; suggest prevention
2	Deadlock prevention, avoidance (banker's algorithm — conceptual), detection, recovery	Apply banker's algorithm to simple example
3	Concurrency: race conditions, critical sections, mutual exclusion, semaphores	Trace concurrent execution; identify race conditions
4	Semaphores and monitors: binary semaphore, counting semaphore; producer-consumer problem	Design semaphore solution for given scenario
5	Consolidation: OS concepts review quiz	Low-stakes retrieval practice

Topic 6.2: Control Systems 控制系统 Kòngzhì Xìtǒng

Week 13 — Sensors, Actuators, and Feedback 传感器、执行器与反馈 Chuángǎnqì, Zhíxíngqì yǔ Fǎnkui

Lesson	Content	Activities
1	Sensors: types (temperature, pressure, light, proximity, motion), analogue vs digital, ADC (传感器, 模数转换器)	Research sensors in smartphones; identify types
2	Actuators: motors, heaters, lights, solenoids; interfaces (DAC, relays, PWM)	Match actuator to control scenario; discuss power requirements
3	Microprocessors vs microcontrollers: architecture, memory, I/O, use cases (Arduino, Raspberry Pi)	Compare specifications; choose device for project
4	Open-loop vs closed-loop control; feedback; set point, error signal, correction	Diagram home heating system; identify loop components

Lesson	Content	Activities
5	TOK: A thermostat "knows" the temperature and "decides" to turn on the heater. In what sense is this knowledge and decision-making? Where is the boundary between control and intelligence?	Discussion; link to AI and embodiment

TOK Knowledge Question: Control systems use feedback to maintain stability. Is this a form of "knowledge" about the environment, or merely a mechanical response? At what point does a feedback loop become sufficiently complex that we might call it "intelligent"?

Week 14 — Control System Applications

Lesson	Content	Activities
1	Industrial control: PLC, SCADA, robotics; manufacturing examples	Case study: automated car assembly line
2	Transportation control: traffic light systems, autopilot, anti-lock braking	Design traffic light control logic; discuss safety
3	Environmental control: smart buildings, irrigation, waste management	Evaluate smart building system for VPA
4	Medical control: pacemakers, insulin pumps, robotic surgery	Ethical discussion: responsibility for control system failures
5	Formative: Control system design exercise (20 marks)	Design feedback loop for given scenario

Week 15 — Distributed and Autonomous Systems Preview (Topic 7 Bridge)

Lesson	Content	Activities
1	Distributed control: multiple controllers, communication, consensus	Research: how do drone swarms coordinate?
2	Autonomous agents: sensors + processing + actuation + decision-making	Analyse self-driving car architecture
3	Reliability and safety: redundancy, fail-safe, fail-soft, watchdog timers	Evaluate reliability of aircraft control systems
4	Topic 6 consolidation: concept mapping across OS and control systems	Collaborative map; identify cross-topic links
5	Formative: Topic 6 progress test (30 marks)	Timed; self-assessment

Week 16 — Topic 6 Deep Revision

Lesson	Content	Activities
1	Revision: Scheduling algorithms — calculate and compare	Problem set; peer checking
2	Revision: Paging, segmentation, virtual memory	Address translation practice
3	Revision: Deadlock and concurrency	Resource graph analysis

Lesson	Content	Activities
4	Revision: Control systems and feedback loops	Diagram construction
5	Exam technique: 10-mark "Explain" and "Evaluate" OS questions	Essay planning; timed writing

Week 17 — Topic 6 Assessment Preparation

Lesson	Content	Activities
1	Mixed revision: Topics 5–6 cross-topic questions	Problem solving
2	Assessment: Topic 6 End-of-Topic Test (1 hour, 50 marks)	Formal conditions
3	Test feedback and DIRT	Markscheme analysis; target setting
4	IA check: all students confirm draft status	One-to-one tutorials
5	Semester 1 examination preparation	Paper 1 practice; exam technique

Week 18 — Semester 1 Examination

Lesson	Content	Activities
1–2	Revision clinic (student-requested topics)	Q&A; problem solving
3	Semester 1 Examination (Paper 1 mock, HL 2h10)	Formal conditions
4	Examination feedback	DIRT; target setting for Topic 7
5	Winter break IA targets set	Individual action plans

Key Vocabulary (Topic 6)

English	中文	Pinyin	Definition
Process	进程	jìnchéng	An instance of a program in execution with its own memory space
Thread	线程	xiàrchéng	A lightweight sub-process sharing memory with other threads
Context switch	上下文切换	shàngxiàwén qiēhuàn	Saving state of one process and loading state of another
Paging	分页	fēnyè	Dividing memory into fixed-size pages for virtual memory
Segmentation	分段	fēnduàn	Dividing memory into variable-sized segments
Deadlock	死锁	sǐsuǒ	A situation where processes are blocked waiting for each other

English	中文	Pinyin	Definition
Semaphore	信号量	xìnhào liàng	A variable controlling access to a shared resource
Sensor	传感器	chuángǎnqì	A device detecting physical properties and converting to signals
Actuator	执行器	zhíxíngqì	A device converting signals into physical action
Feedback loop	反馈回路	fǎnkuì huílù	A system where output influences future input
Microcontroller	微控制器	wēi kòngzhì qì	A compact integrated circuit designed for embedded control

Teacher Checklist — End of Topic 6:

- ☐ All IB 6.1 – 6.2 syllabus points covered
- ☐ Students can calculate scheduling metrics
- ☐ Students can translate logical to physical addresses
- ☐ Students can identify and resolve deadlock
- ☐ Students can design feedback control systems
- ☐ IA first drafts reviewed and feedback given
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 6 Scheme of Work